

CRISPR in action: practice and theory of
genome engineering

Day 5: Beyond CRISPR-Cas9 and gene editing

Ruben Vazquez Uribe
August 2026



Why CRISPR is so important and revolutionary?



PRECISION



SIMPLICITY



BROAD
APPLICATIONS



AFFORDABILITY



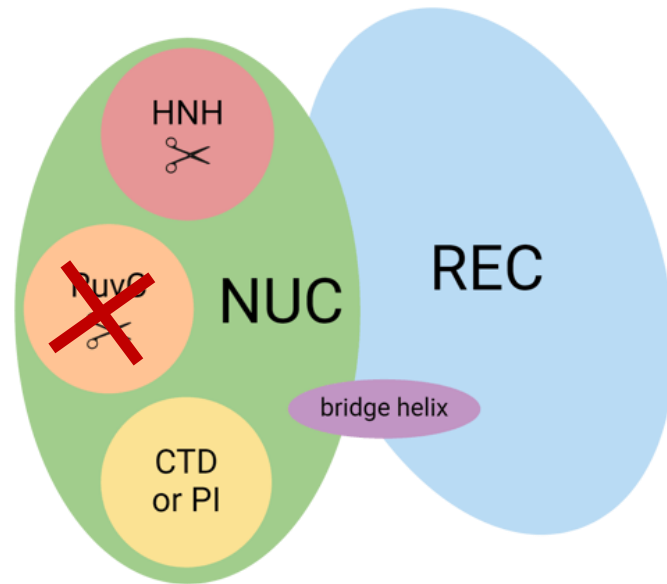
SPEED



POTENTIAL



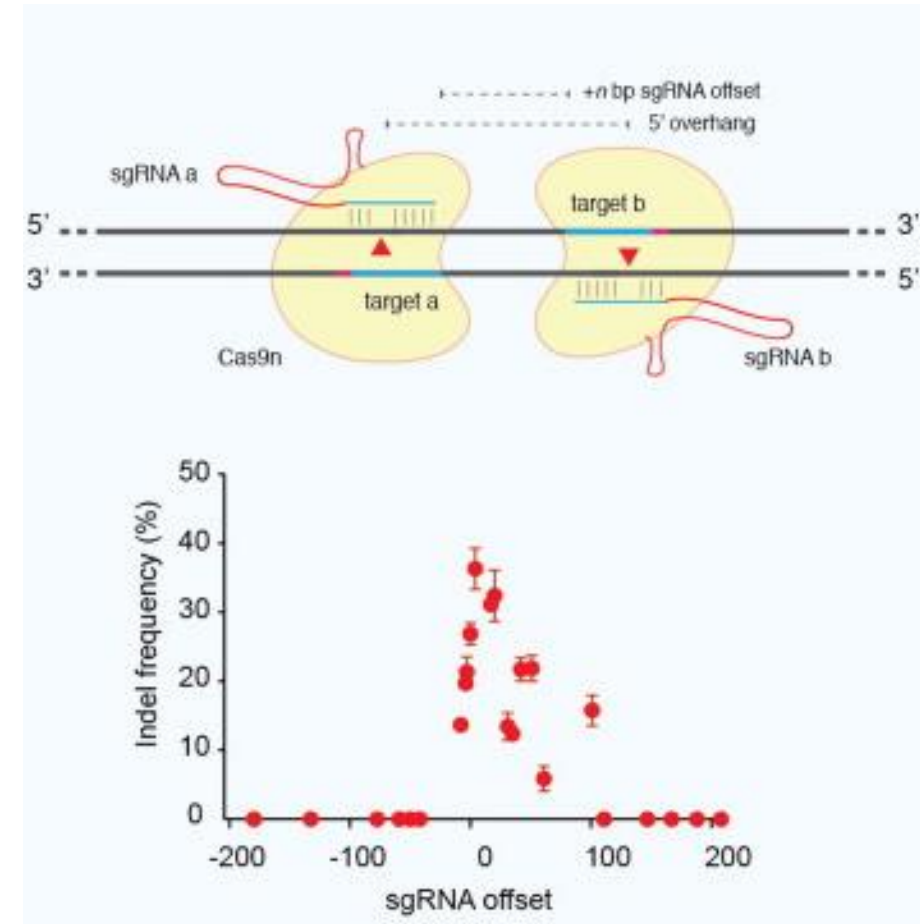
We are here

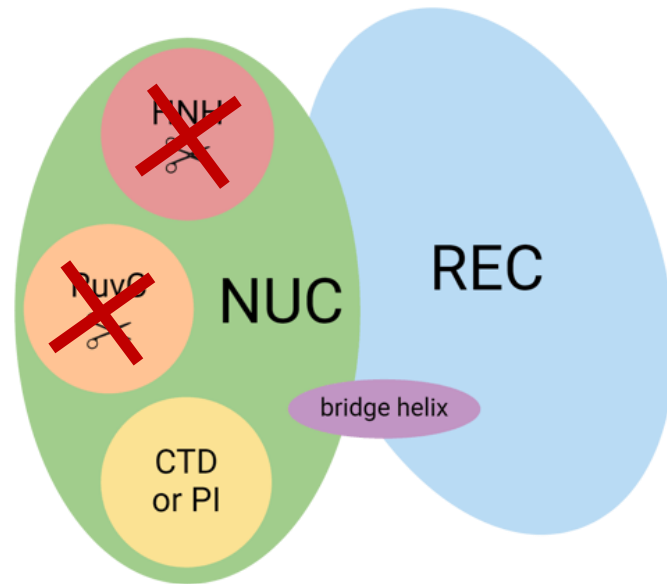


Cas9 → Cas9n (= cas9 nickase)

Cas9 nickases reduced off targets compared to Cas9

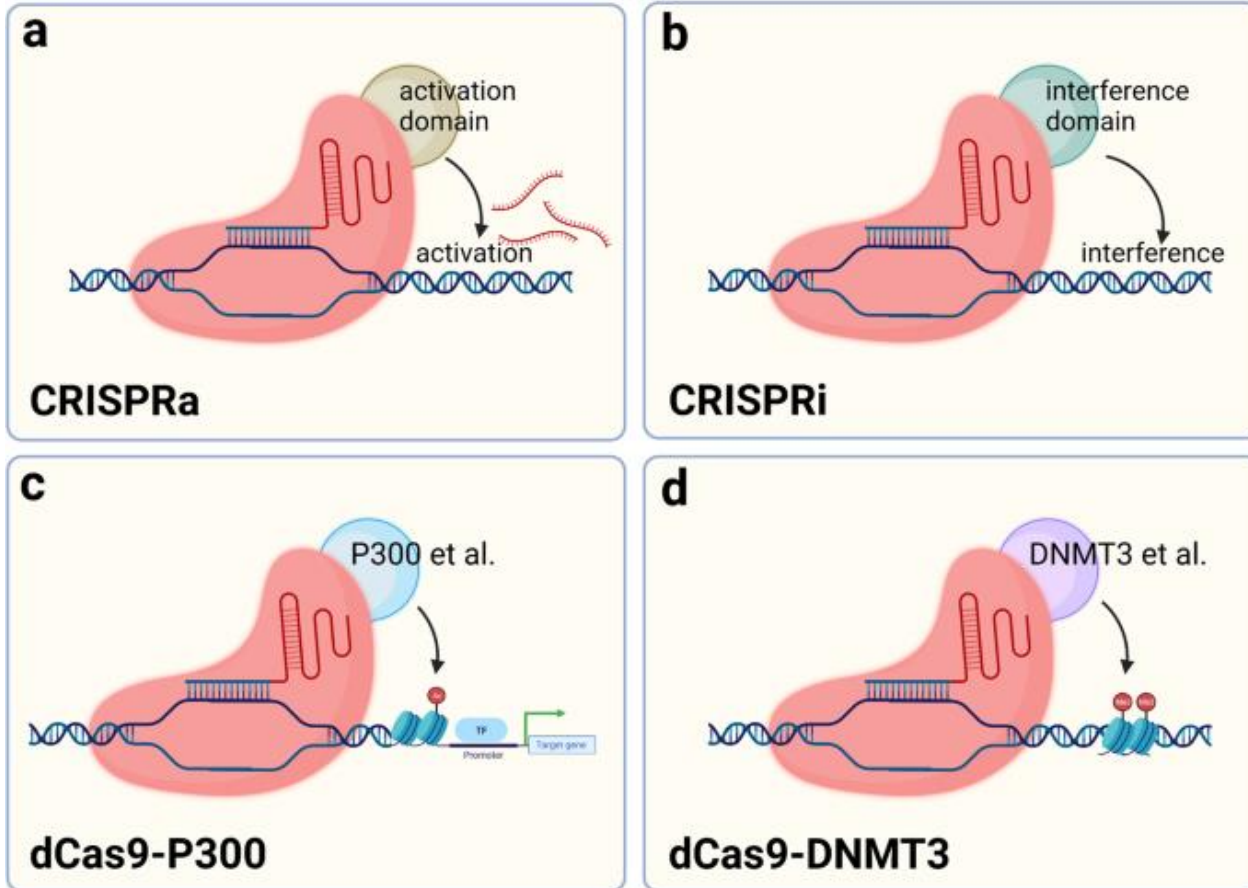
- Cas9 nickase can facilitate targeted DNA double-strand break using two guide RNAs
- Double nicking of DNA reduces off-target mutagenesis by 50- to 1,000-fold
- Multiplex nicking stimulates homology directed repair, microdeletion, and insertion
- Paired nicks can produce overhanging DNA ends that are sometimes more suitable for precise gene insertion or correction by HDR, boosting the accuracy of inserting or repairing genes





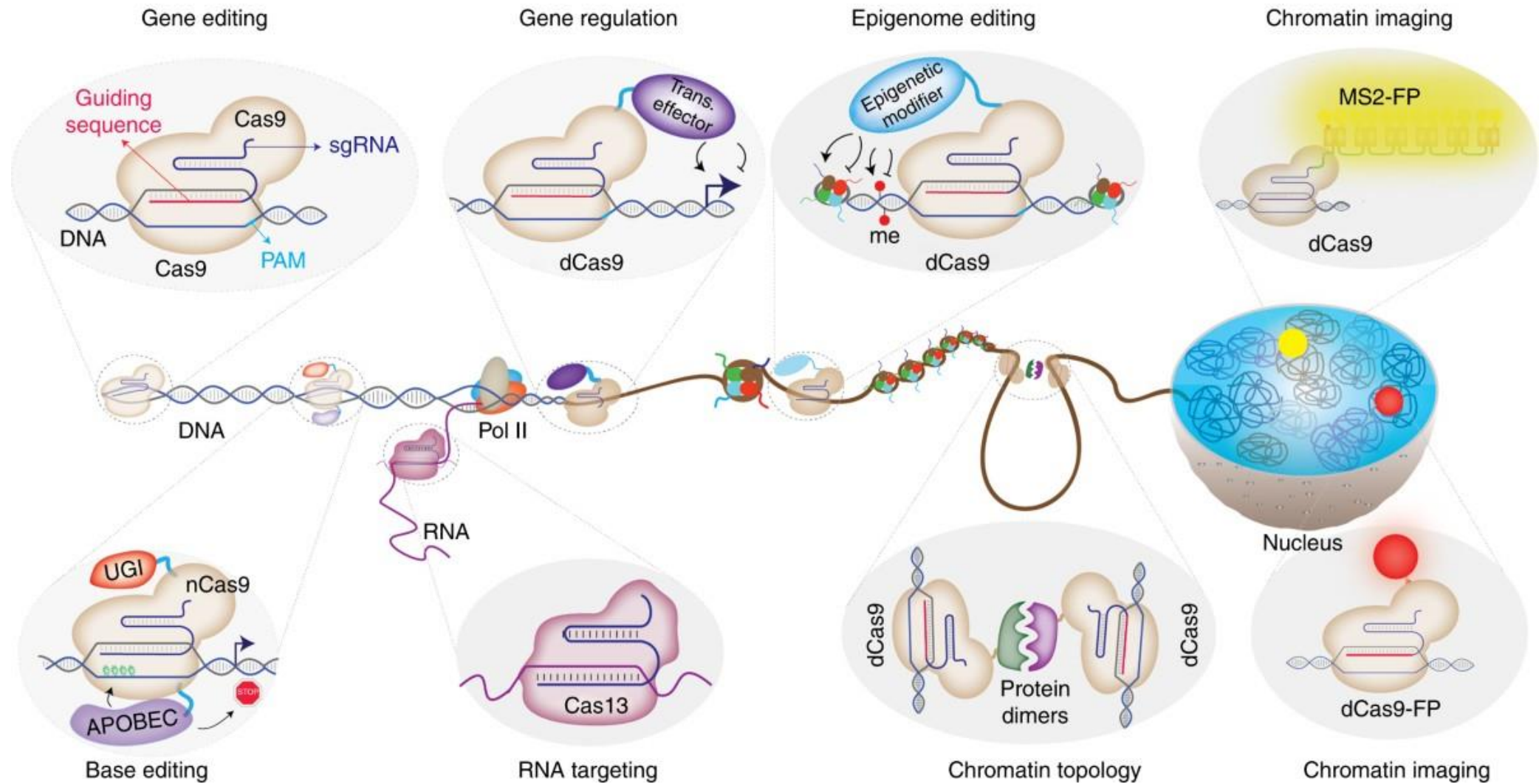
Cas9 → dCas9 (=dead cas9)

dCas9-based tools to regulate expression

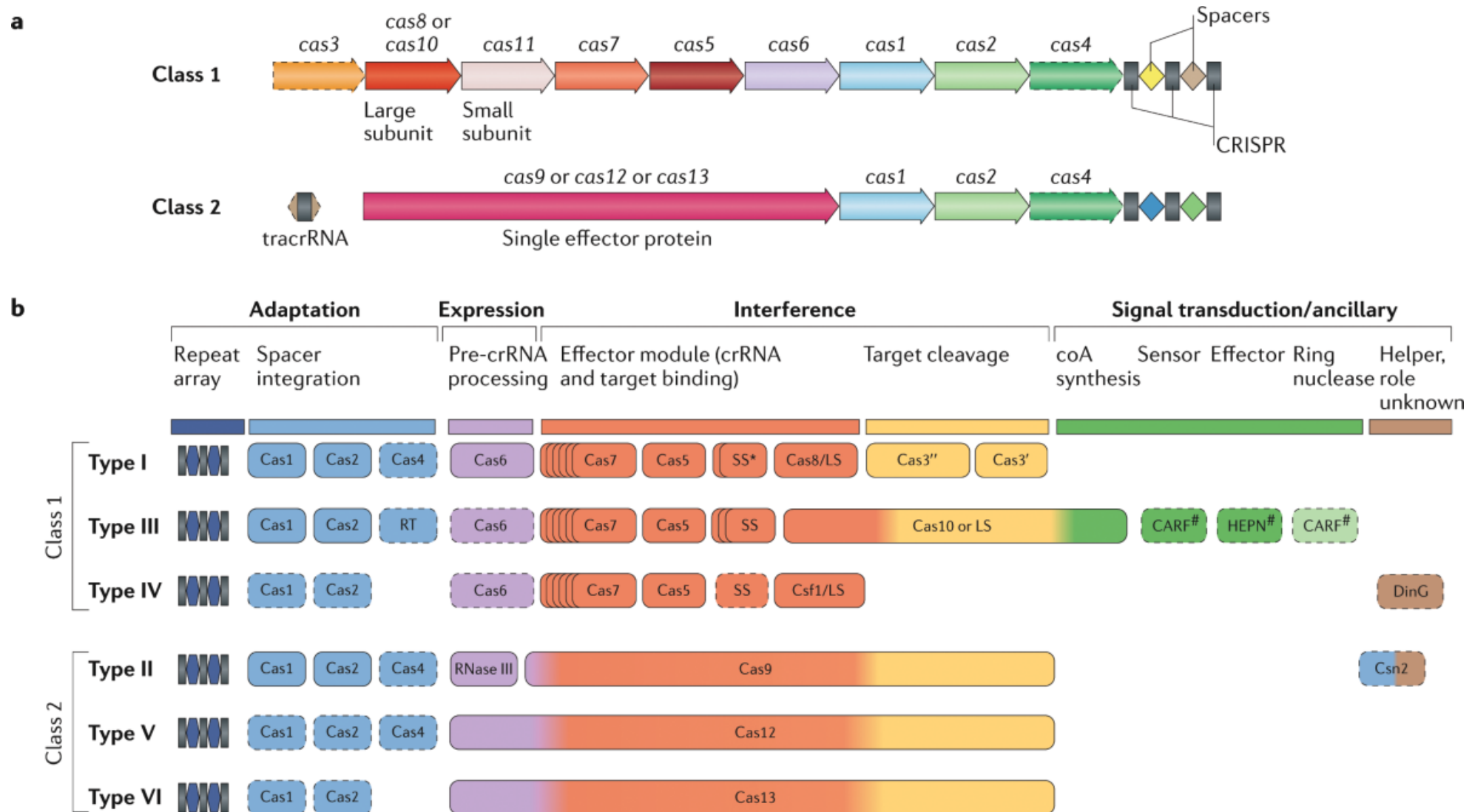


- a. The dCas9 fusion to transcriptional activator to promote gene transcription
- b. dCas9 may be fused to transcriptional repressor effectors to inhibit gene transcription
- c. the fusion of dCas9 with histone acetylases promotes histone acetylation near the promoter region, which in turn enhances gene transcription
- d. fusion of dCas9 with DNA methyltransferases to knock down gene transcription

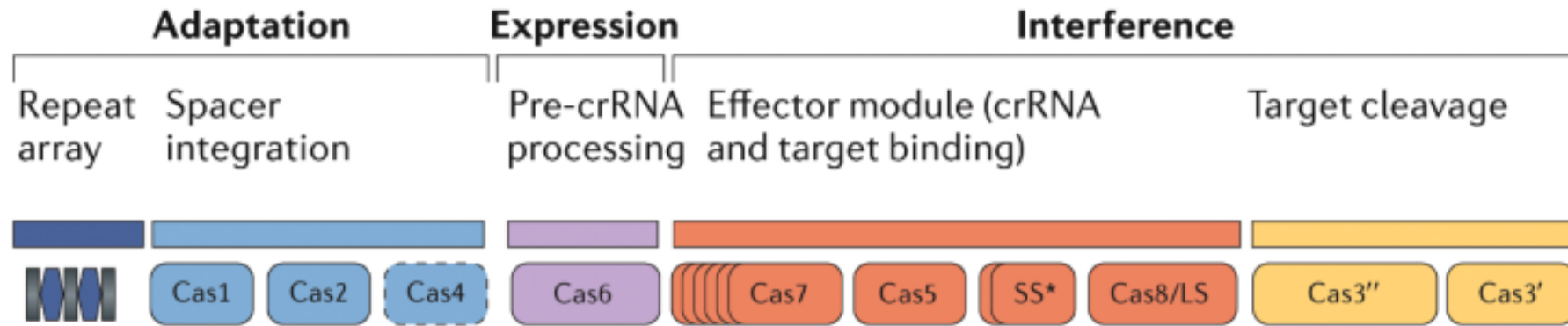
CRISPR technology: Beyond genome editing



Beyond Cas9...

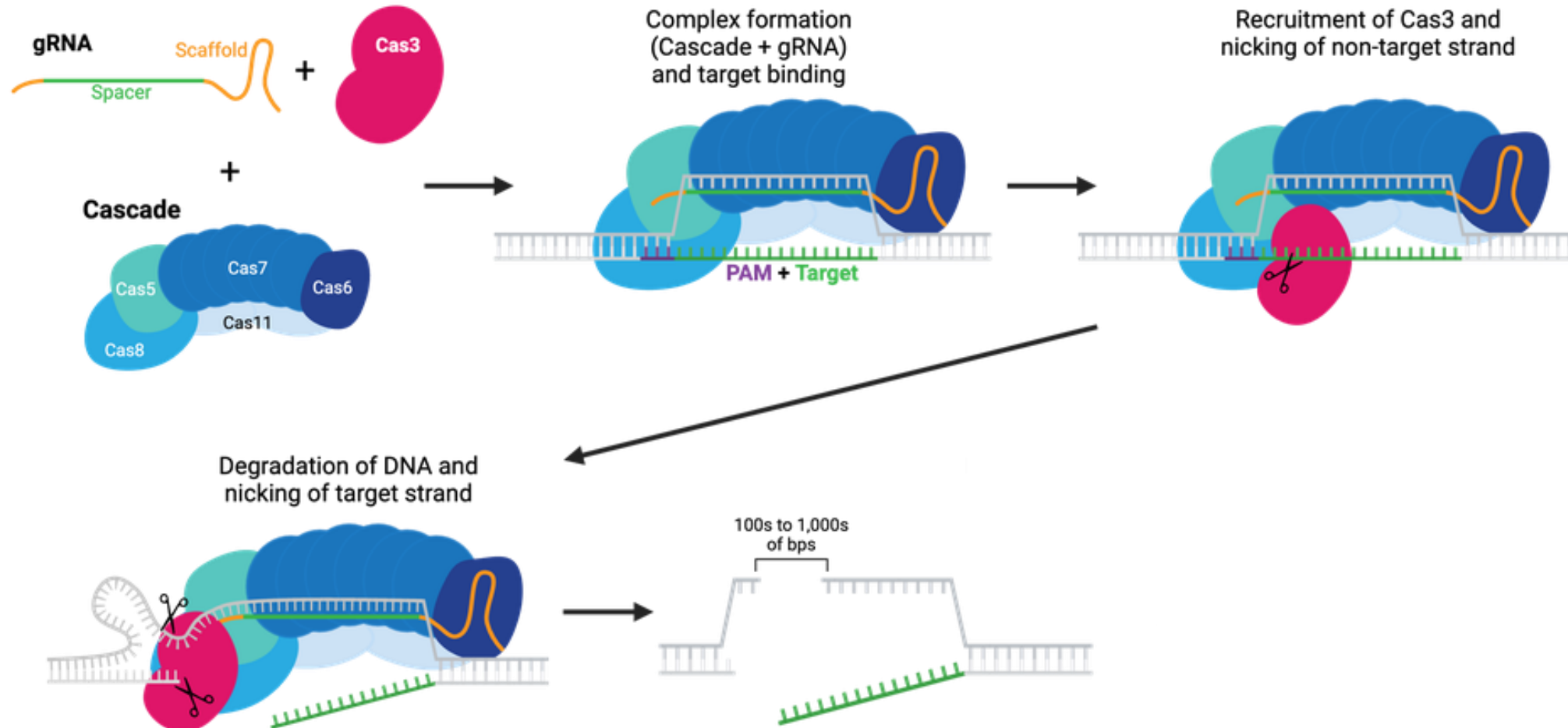


Type I CRISPR-Cas systems



- Most common type of CRISPR overall, especially prominent in *Actinobacteria*, *Proteobacteria*, and *Firmicutes*.
- Cas3 is the effector protein in this system. It is an ATP-dependent single-stranded DNA (ssDNA) nuclease and helicase.
- Type I systems are used in gene editing primarily for large-scale, directional DNA deletions rather than the small, precise cuts typical of Cas9.

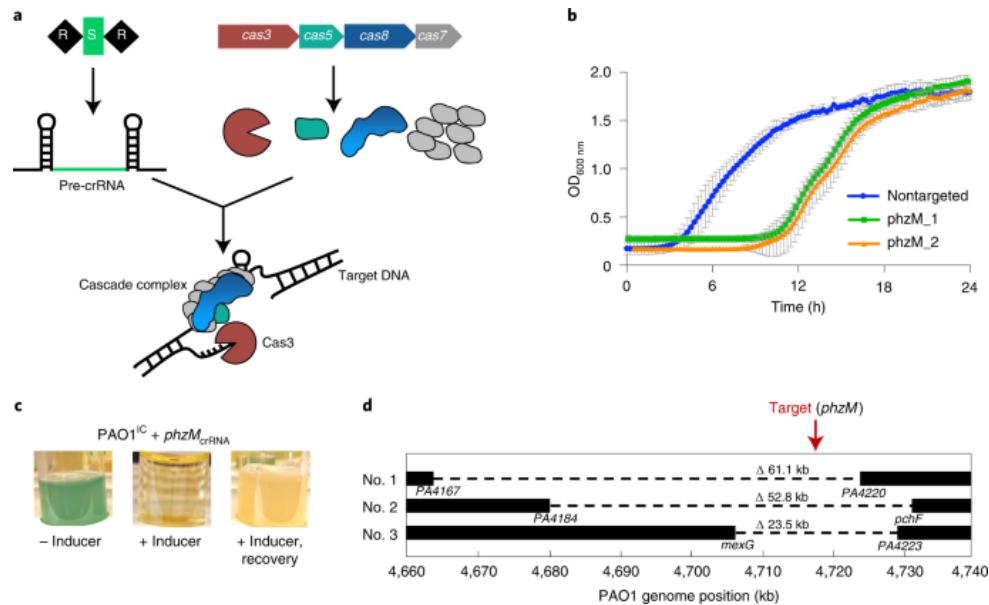
Type I CRISPR-Cas systems



Large genomic deletions in prokaryotes and eukaryotes

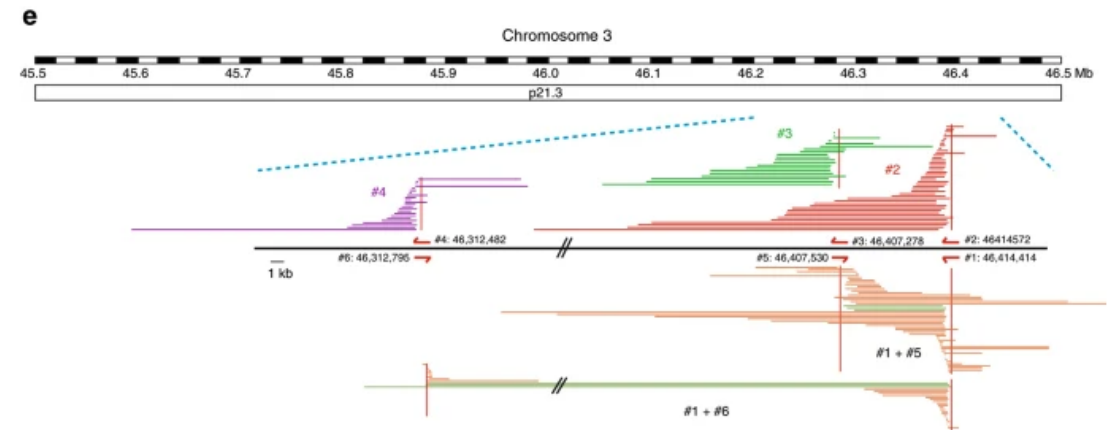
A compact Cascade-Cas3 system for targeted genome engineering

Bálint Csörgő^{1,2,3,9}, Lina M. León^{1,3,9}, Ilea J. Chau-Ly^{1,4}, Alejandro Vasquez-Rifo^{1,5}, Joel D. Berry¹, Caroline Mahendra^{1,5}, Emily D. Crawford^{1,6}, Jennifer D. Lewis^{1,7} and Joseph Bondy-Denomy^{1,3,8}✉



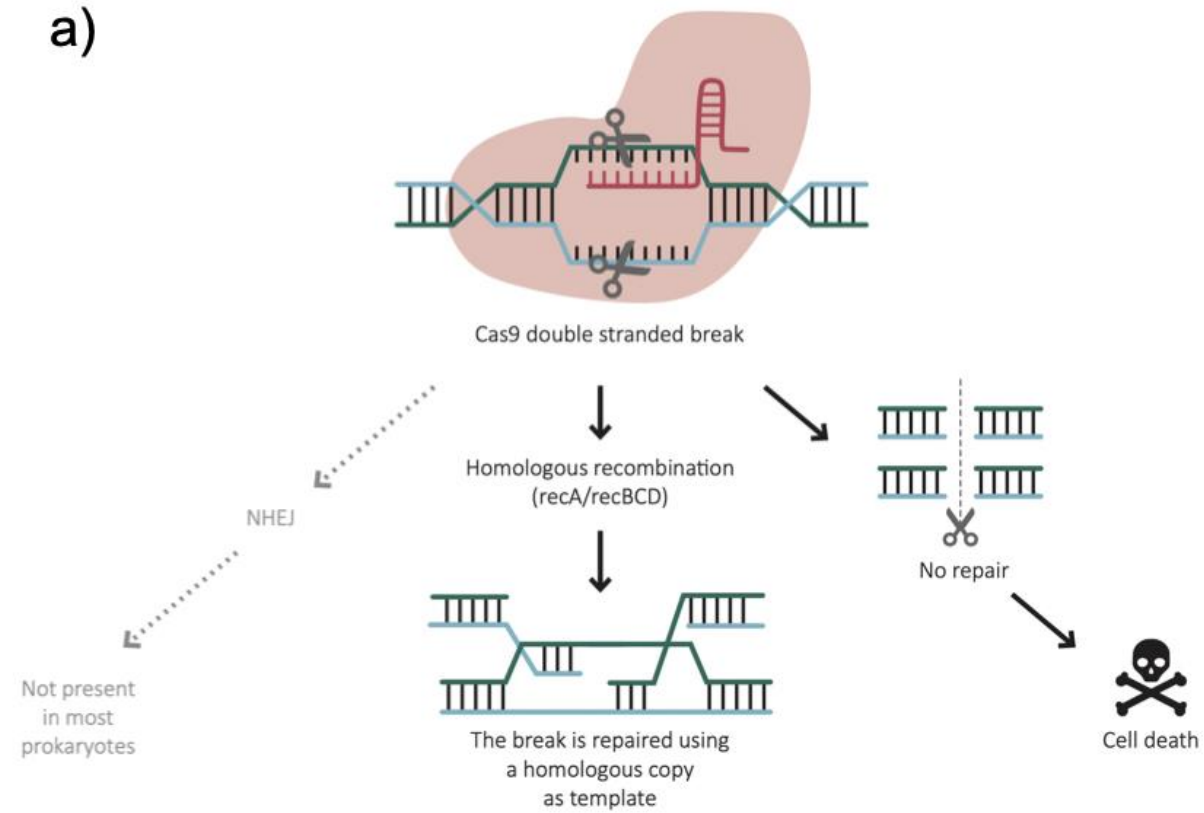
CRISPR-Cas3 induces broad and unidirectional genome editing in human cells

Hiroyuki Morisaka^{1,2,13}, Kazuto Yoshimi^{3,4,12,13}, Yuya Okuzaki^{1,5,13}, Peter Gee^{1,5}, Yayoi Kunihiro^{1,3}, Ekasit Sonpho^{4,6}, Huaigeng Xu^{1,5}, Noriko Sasakawa^{1,5}, Yuki Naito^{7,8}, Shinichiro Nakada⁹, Takashi Yamamoto¹⁰, Shigetoshi Sano², Akitsu Hotta^{1,5*}, Junji Takeda^{1,11*} & Tomoji Mashimo^{3,4,12*}

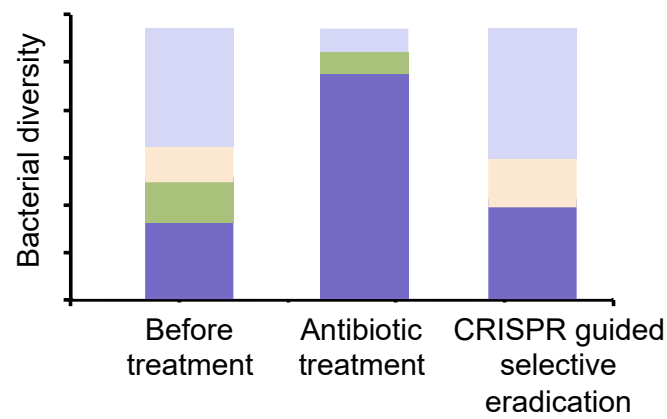


CRISPR antimicrobials

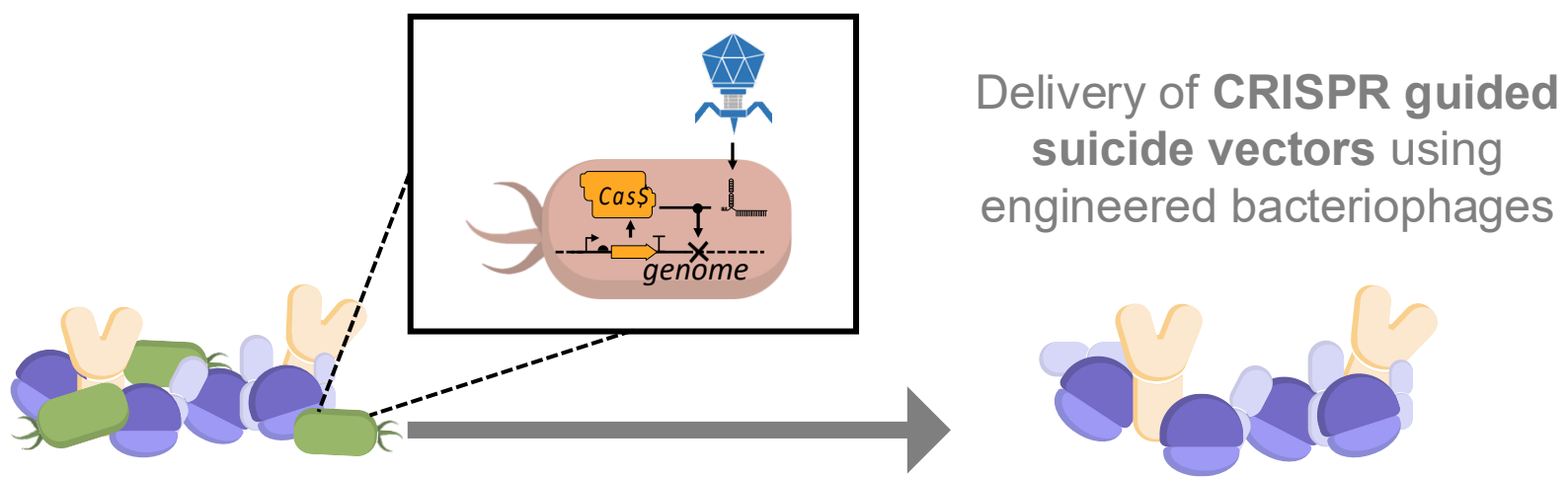
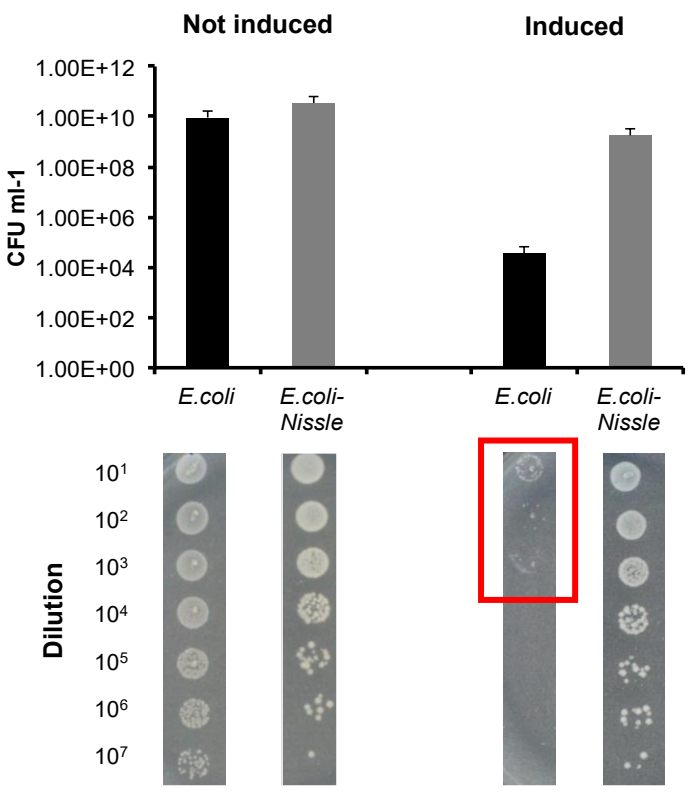
a)



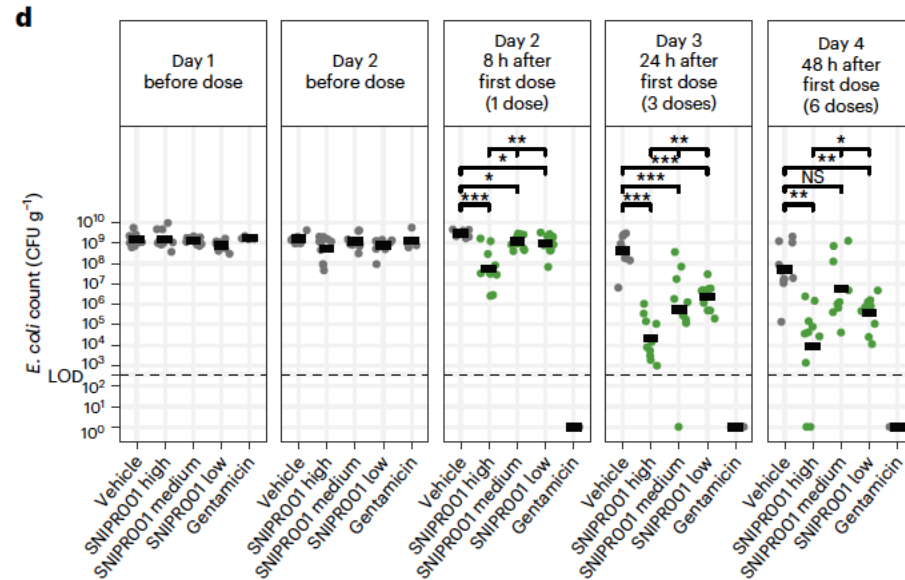
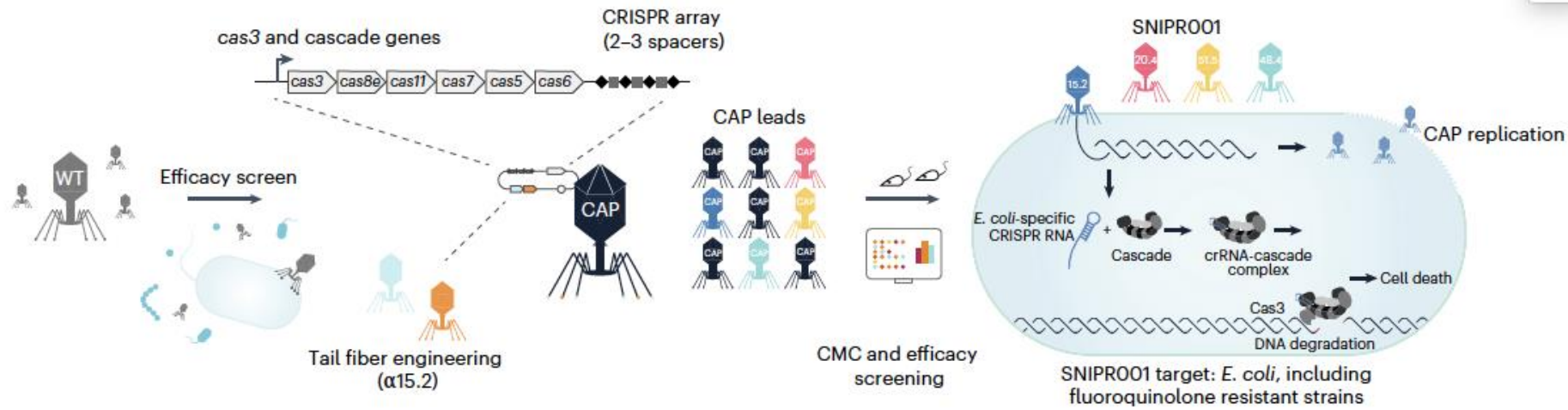
CRISPR antimicrobials



This technology shall enable programmable eradication of specific species by **harnessing diverse Cas systems.**



CRISPR antimicrobials



E. coli recovery in mouse feces indicates increased SNIPR001 effect with increased dose.

CRISPR antimicrobials

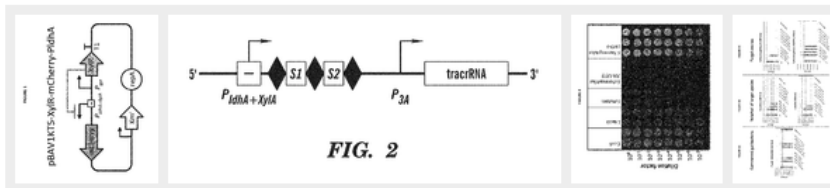


Altering microbial populations and modifying microbiota

Abstract

The invention relates to methods, uses, systems, arrays, engineered nucleotide sequences and vectors for inhibiting bacterial population growth or for altering the relative ratio of sub-populations of first and second bacteria in a mixed population of bacteria. The invention is particularly useful, for example, for treatment of microbes such as for environmental, medical, food and beverage use. The invention relates inter alia to methods of controlling microbiologically influenced corrosion (MIC) or biofouling of a substrate or fluid in an industrial or domestic system.

Images (12)



US9701964B2

United States

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Inventor: Jasper Clube, Morten Sommer, Christian Grøndahl, Eric VAN DER HELM, Ruben VAZQUEZ-URIBE

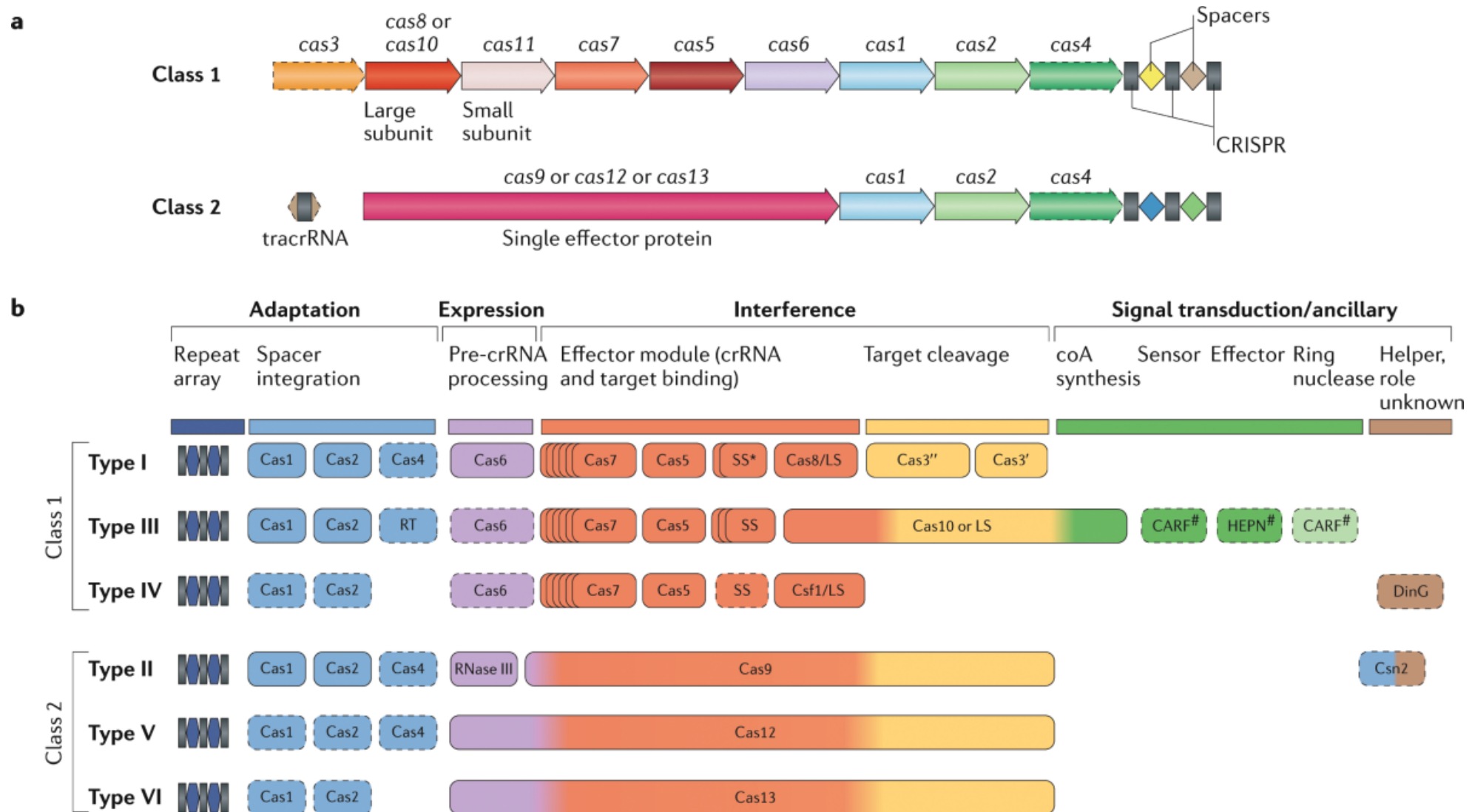
Current Assignee: SNIPR Technologies Ltd

Worldwide applications

2016 • [HK](#) [EP](#) [EP](#) [KR](#) [WO](#) [SI](#) [EP](#) [IL](#) [EP](#) [EP](#) [EP](#) [KR](#) [CN](#) [DE](#) [PL](#) [IL](#) [EP](#) [NZ](#) [EP](#) [KR](#) [KR](#) [EP](#) [CN](#) [EP](#) [IL](#) [CA](#) [EP](#) [AU](#) [ES](#) [RU](#) [JP](#) [DK](#) [EP](#) [SG](#) [US](#)
2017 • [US](#) [US](#) [IL](#) [US](#) [US](#) [US](#) [US](#) [US](#) 2018 • [US](#) [US](#) 2019 • [US](#) [AU](#)
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- **A Phase 1a trial** in healthy volunteers, where CRISPR antimicrobials demonstrated promising safety and target engagement.
- An ongoing **Phase 1b/2a** randomized, double-blind, placebo-controlled clinical trial (NCT06938867) in patients with hematological cancer undergoing hematopoietic stem-cell transplantation.

Beyond Cas9...



Type V CRISPR-Cas systems

- Cas12 = Cpf1
- Cas12 is an RNA-guided enzyme.
- Targets dsDNA like Cas9
- After cleavage, Cas12a remains active and can exhibit collateral cleavage activity, indiscriminately cutting single-stranded DNA



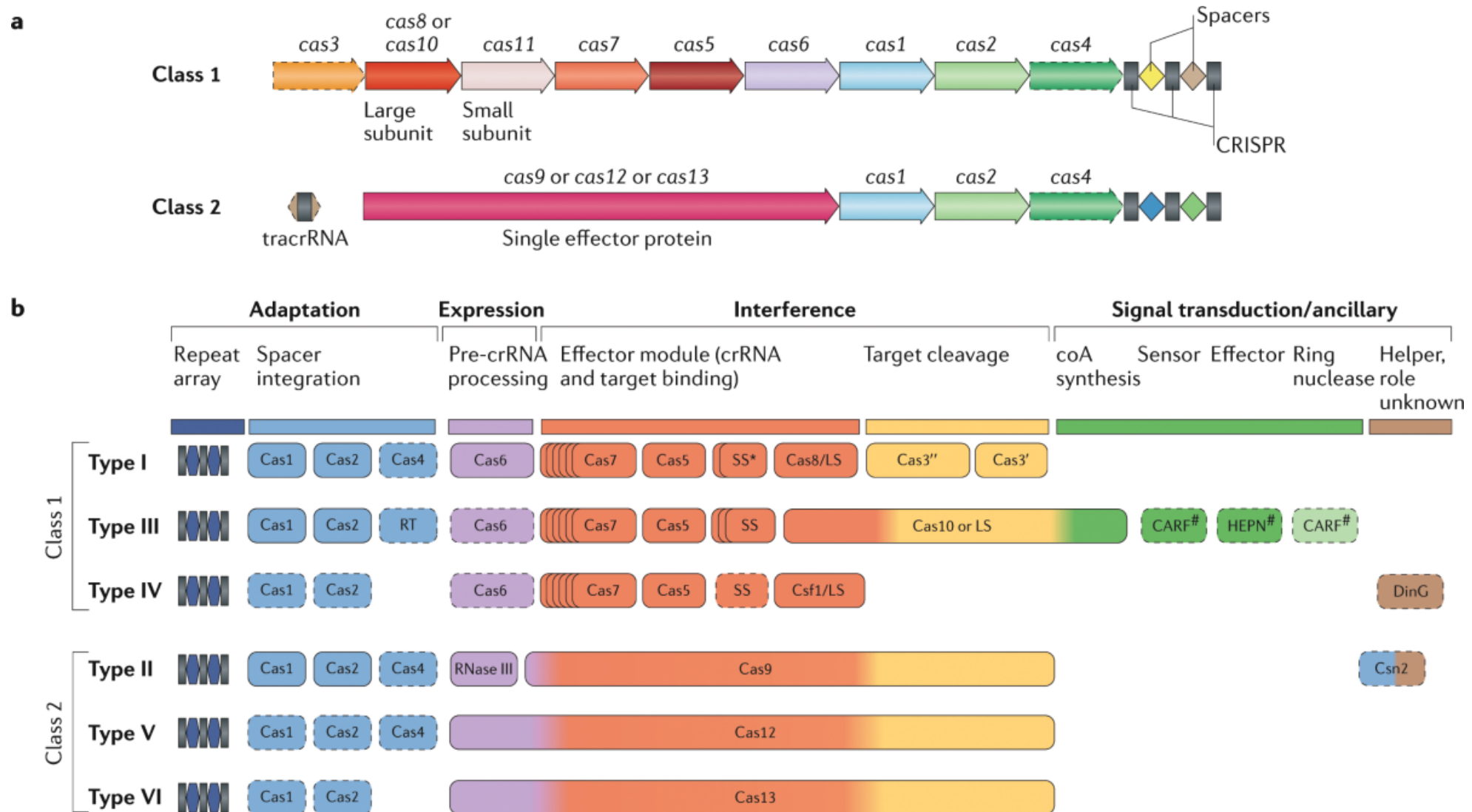
Aspect	SpCas9	LbCas12 (Cas12a)
DNA Cut Type	Blunt	Staggered/sticky ends
PAM	NGG (GC-rich)*	TTTV (AT-rich)*
Multiplex Editing	Complex (needs tracrRNA)	Easy (single crRNA)
Off-target Effects	Can be higher ¹	Generally lower ²
Collateral Activity	No	Possible on ssDNA
Protocol Maturity	Very high	Less mature

*The PAM is different in other Cas9 and Cas12 orthologs

¹ Jinek et al., 2012

² Paul and Montoya, 2020

Beyond Cas9...



Summary of CRISPR-Cas systems

CRISPR Type	Signature Effector(s)	Main Target
Type I	Cas3	dsDNA
Type II	Cas9	dsDNA
Type III	Cas10, Csm/Cmr complex	RNA and DNA
Type IV	Csx protein families	Likely DNA (exact unclear)
Type V	Cas12, Cas14	DNA (dsDNA and ssDNA)
Type VI	Cas13	RNA

Article

Conversion of IscB and Cas9 into RNA-guided RNA editors

Chengtao Xu,¹ Xiaolin Niu,¹ Haifeng Sun,¹ Hao Yan,^{2,3} Weixin Tang,^{2,3} and Ailong Ke^{1,4,*}

¹Department of Molecular Biophysics and Biochemistry, Yale University, New Haven, CT 06512, USA

²Department of Chemistry, The University of Chicago, Chicago, IL 60637, USA

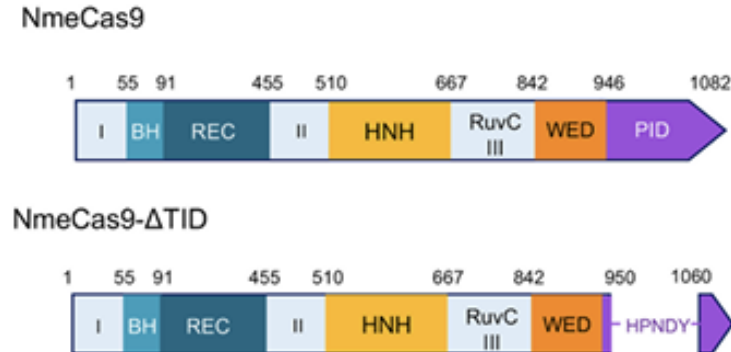
³Institute for Biophysical Dynamics, The University of Chicago, Chicago, IL 60637, USA

⁴Lead contact

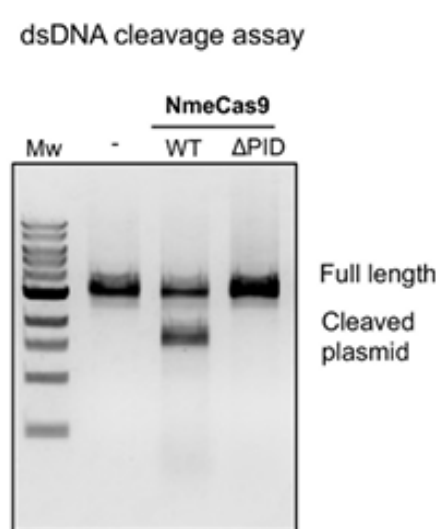
*Correspondence: ailong.ke@yale.edu

<https://doi.org/10.1016/j.cell.2025.07.032>

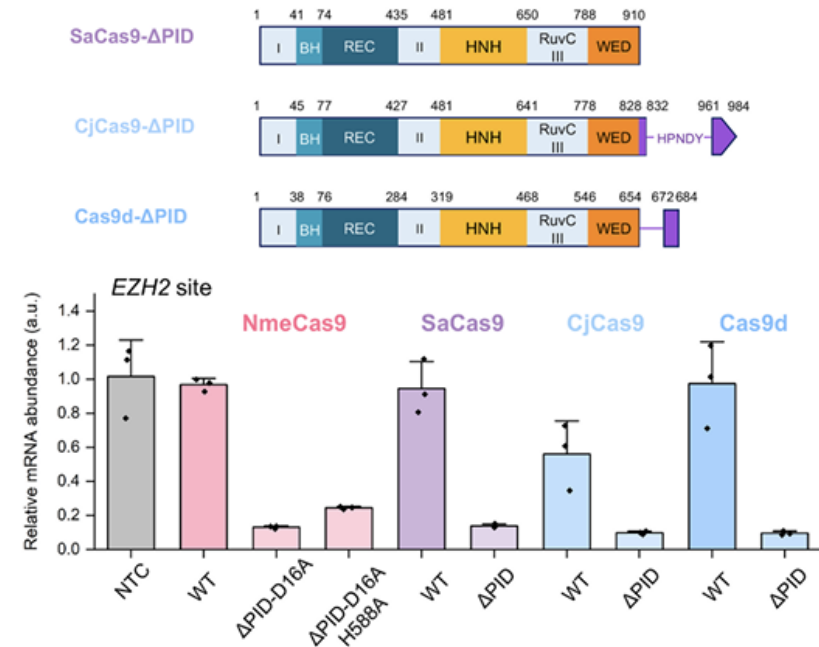
A



B

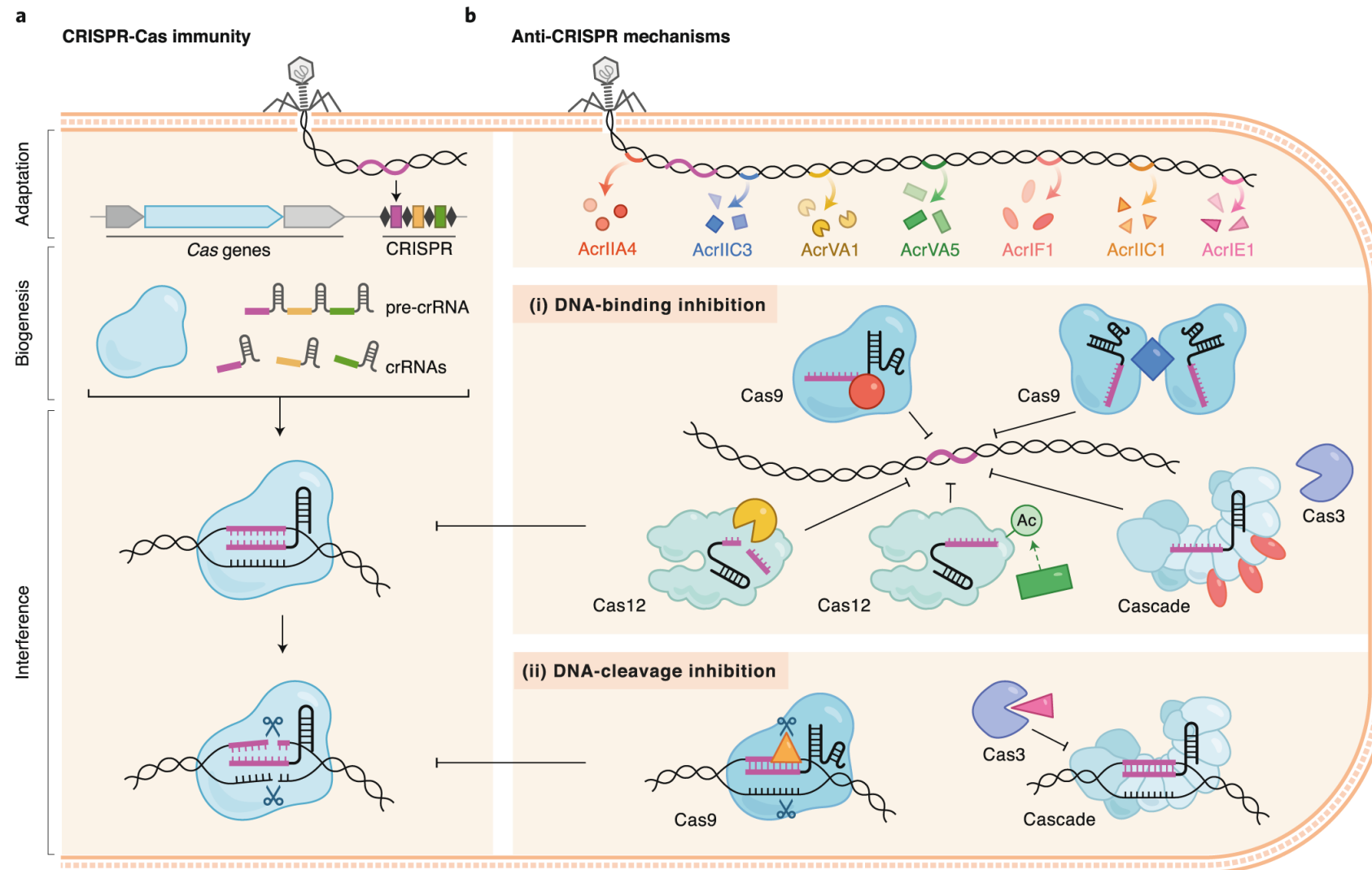


D

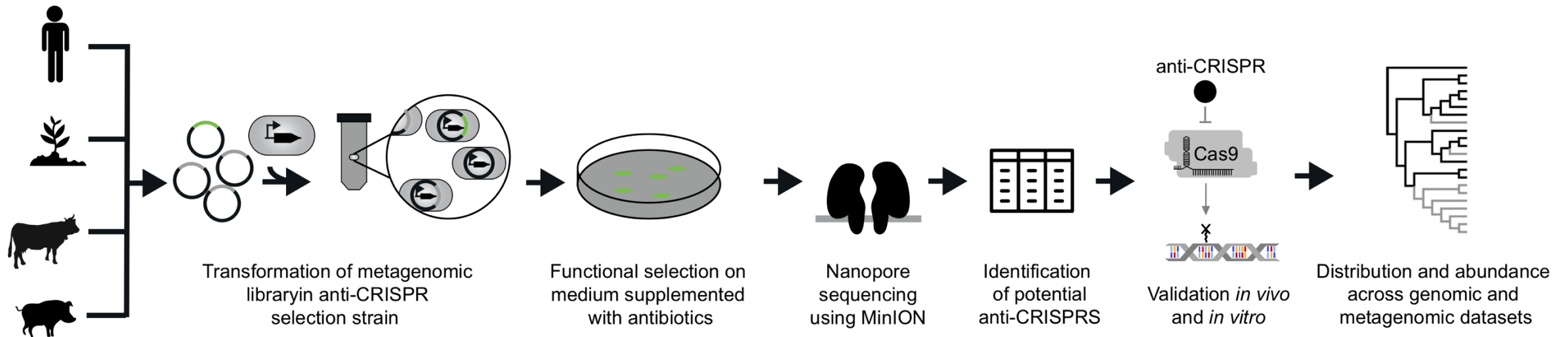


There is even more beyond CRISPR...

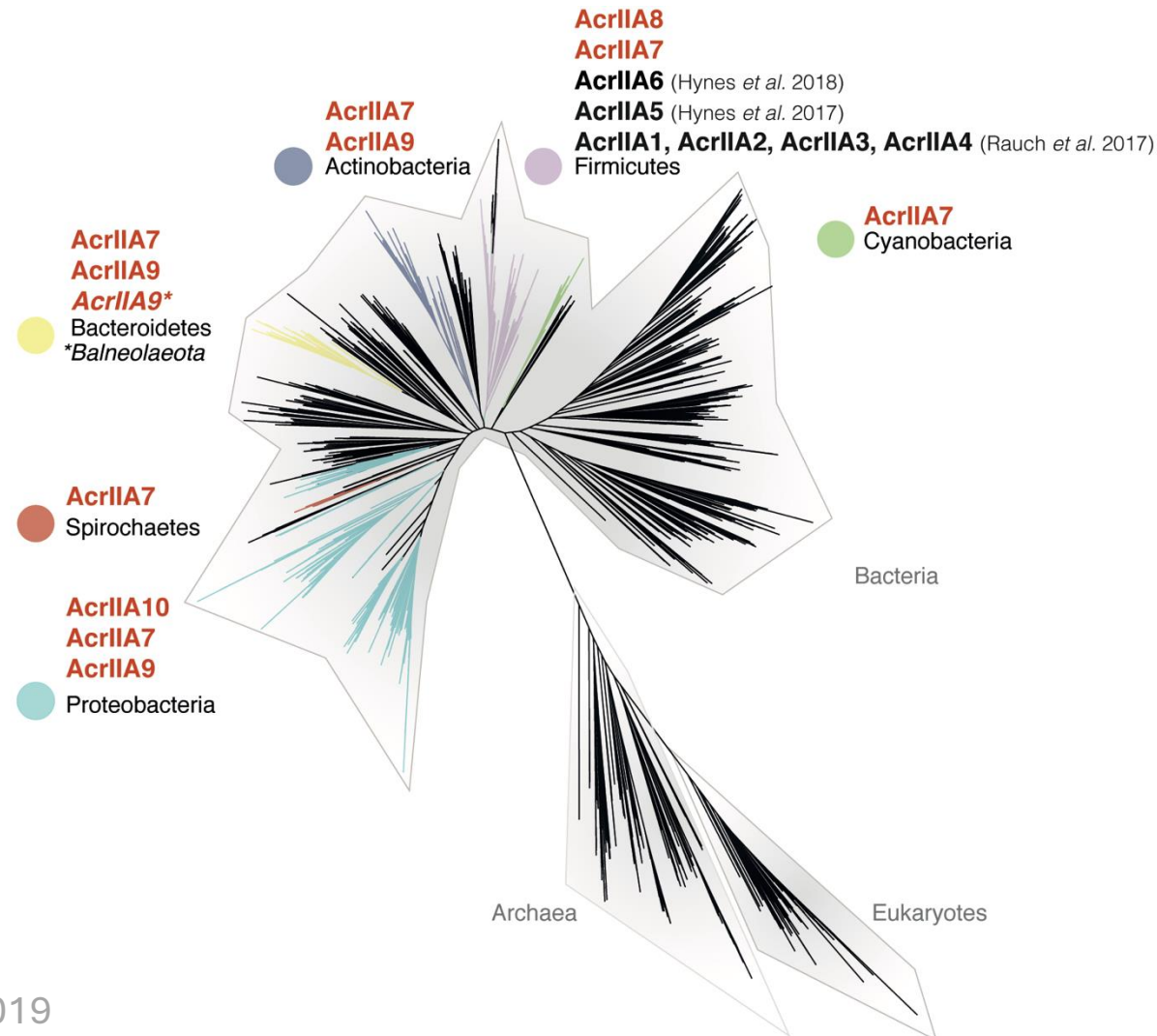
Anti-CRISPRs



Workflow for identification of ACR proteins from metagenomic libraries

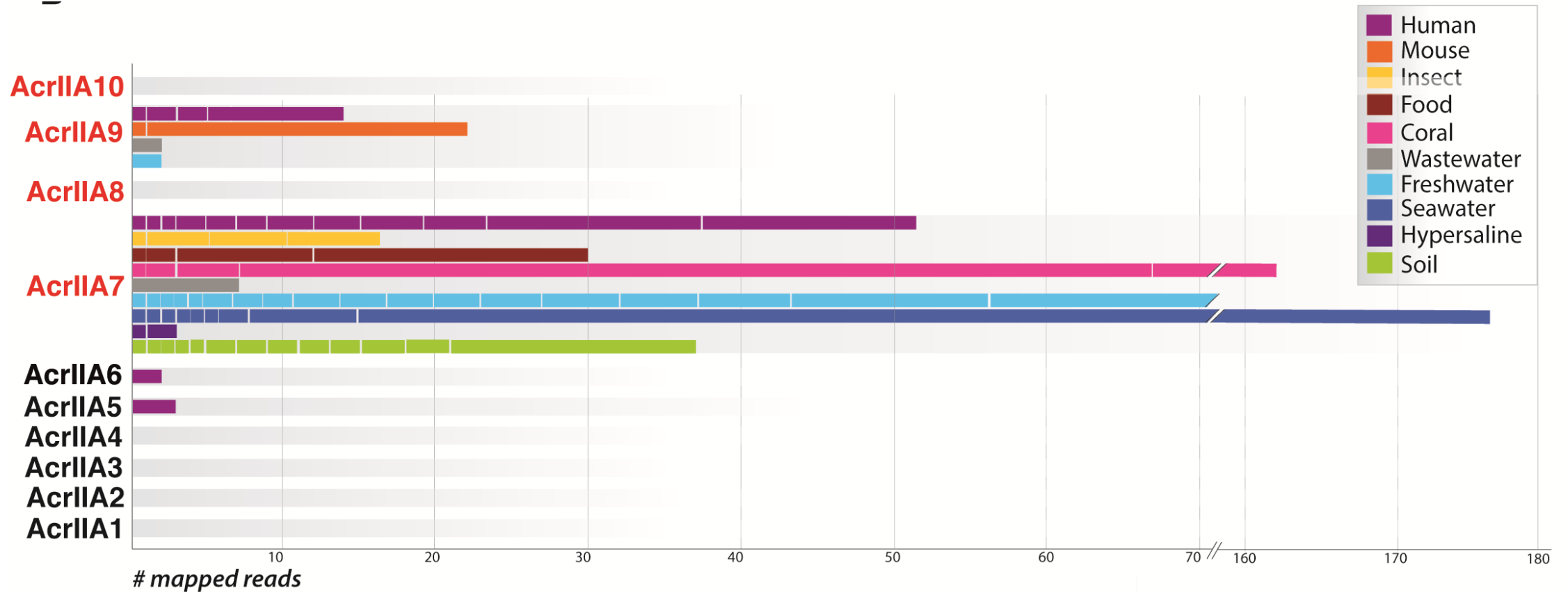


Distribution of AcrIIAs across genomic and metagenomic datasets



Hosts of all AcrIIAs mapped to the tree of life (Hug *et al.*, 2016)

Distribution of AcrIIAs across genomic and metagenomic datasets



Applications of anti-CRISPRs

- **“Off-Switch” in Gene Editing:** Spatial or temporal control of edits.
- **Biosafety:** Anti-CRISPRs offer a tool to limit the spread of engineered gene drives in wild populations, providing a biosafety strategy for environmental or pest control uses.
- **Biocontainment:** as programmable synthetic toxin-antitoxin systems in bacteria (Choudhary et al., 2023)
- **Enhanced phage therapy:** Since anti-CRISPRs are natural phage proteins, they can be used to help therapeutic phages overcome bacterial CRISPR defenses.

Take home messages

- Cas9 is powerful, but it's not the whole story.
- CRISPR-Cas systems come in many flavors, each with unique potential beyond genome editing.
- Consider the right CRISPR tool for your application. But remember that alternatives may be less well characterized.